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Report on the IEEE Region 10 Conference 2024 (IEEE TENCON 2024, https://tencon2024.org/)

Conference

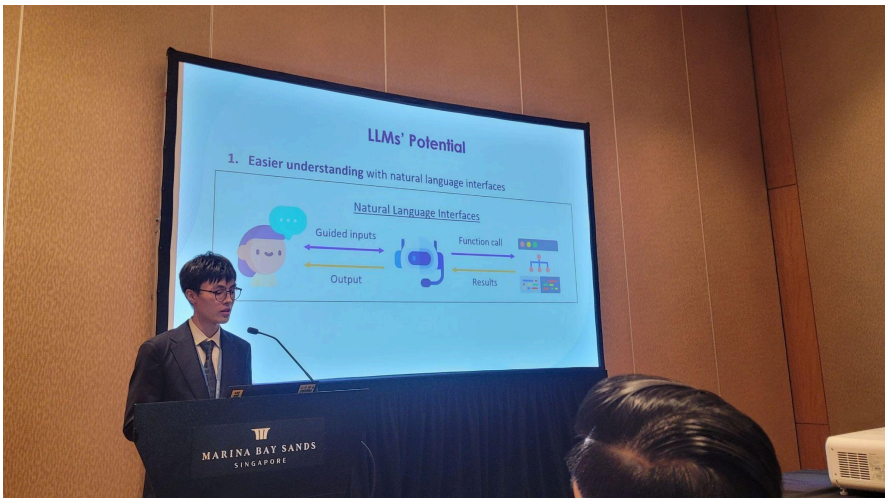
The IEEE Region 10 Conference 2024 (IEEE TENCON 2024) was held from 1 December 2024 to 4 December 2024 at Marina Bay Sands Singapore, with the theme 'Artificial Intelligence and Deep Learning Technologies for a Sustainable Future'. All works presented at this conference were heavily focused on the topics of Artificial Intelligence (AI), its development, and its applications. The sub-themes presented include Communications, Electronics, Medical Applications, Power and Energy, and Other Emerging Technologies. This conference heavily focused on the application of AI in these fields and the works presented many applications in novel use cases and scenarios. This allows the power of AI to be harnessed for positive applications that achieve the goal of a sustainable future.

Contributions

I am the author and presenter of the paper titled 'Knowledge-Based Extraction for Inverse Design of Optical Thin-Films'.

Optical thin-films consist of nanometre-thin materials deposited on a substrate, designed to have specific optical properties to serve a variety of use cases. However, due to the challenges associated with their optimisation due to the presence of many parameters, we propose a Large Language Models (LLMs)-based tool. Being LLM powered, this creates an integrated platform for users to perform various functions related to inverse design of thin-films using an accessible natural language interface. The knowledge extraction of information also helps to provide clear, concise, and expert answers to questions, greatly enhancing the design knowledge of optical thin-films. This developed platform significantly reduces the resources and time for designers in developing optical thin-films to reach an optimal design. This will allow thin-films' applications to be brought to the market faster and with increased efficiency.

We developed this work and I orally presented this work at IEEE TENCON 2024 on 2 December 2024 for a time slot of 15 minutes at a breakout session, inclusive of time to address questions. Multiple questions were also asked, including clarifications such as whether we trained a model and questions regarding future directions such as the exploration of other LLMs.



Observations

This conference surfaced many novel use cases of AI and evaluated their accuracy, viability and potential to be further extended.

Research in Developing Economies

Since the focus was on the use of AI for a sustainable future, the consideration of access to AI technologies, especially in countries that are less developed, was an issue that was highlighted. This is as the sustainable aspect of development can only be achieved if it is inclusive. Borines et al. presented a literature review that analysed the limitations of AI in emerging economies (which was defined in the work to be countries of Human Development Index below the top 57 countries).[1] It highlighted how even though AI might be being applied, skill and data management shortages still exist, posing a big barrier to its use.

Similarly, there were also works that presented novel applications of AI for these residents of less developed countries. Silapachote et al. presented the use of YOLOv5, a computer vision model, to identify poultry diseases from images of chicken feces.[2] This was then implemented into the LINE messaging app to allow easy use by many farmers as a more efficient and accurate way to identify these diseases.

Bautista et al. also presented a work that uses Deep Neural Networks to detect illegal electricity pilferage (e.g. meter bypass, diverted load, illegal tapping) based on estimating the expected electrical meter reading and comparing it to the actual meter reading.[3] This allowed detection and localisation of these incidents of illegal electricity modification which is common is these developing economies.

Novel Applications of AI

Many of these works also heavily focus upon the specific applications of AI and possible challenges in these niches. For example, Ercan and Mathyalakan proposed the use of computer vision based inspection in Singapore's public housing to detect clutter.[4] Corridors have a minimum clear width requirement for safety reasons. They implemented a robot equipped with LIDAR to detect said width and computer vision to detect the types of clutter that violated these rules (e.g. shoe racks, bicycles, plants). They found that this model of detecting specific categories of items, as opposed to a binary model predicting clutter or uncluttered, was much more accurate.

In another work, Abudin et al. were able to use data interpolation to recover lost data due to false data injection attacks in smart grids.[5] Smart grid technology is often used to better distribute power resources and optimise generation and transmission, but it is subject to data attacks on the systems that inform the allocation of resources. Recovering lost data is also very time intensive. As such, they propose the use of spline interpolation as a way to reconstruct damaged data, especially as these attacks tend to be over short durations. This allows them to ensure the efficacy of such smart grids.

Novel Architectures

Yoshida et al. presented on using a GAN-based tone curve approach to construct a High Dynamic Range (HDR) image from one Low Dynamic Range (LDR) image.[6] Traditional methods of rendering require multiple LDR images at different exposures to render one HDR image. However, using only one LDR image, they propose the use of generative AI to generate two more LDR images, one up exposed and one down exposed, to be rendered to create the HDR image. Most GANs typically operate as black boxes with difficulty in understanding their operation. Hence, the use of a tone curve approach allows better and easier understanding of the model for more effective prediction as well.

Paul et al. also developed a slightly more novel architecture of using densified causal knowledge graphs (CKG) to identify cyber attacks.[7] By using informative CKGs, they can address complex interconnections of events in industrial cybersecurity. Using cause-effect relationships, this allows them to map out such events to better identify cyber attacks. They were also able to develop a metric to evaluate the performance of the CKG based on entropy, and it is a novel structure and framework for this application.

Overall

In general, this conference presents many different aspects and applications of AI and truly demonstrates the versatility of AI in our lives. There are many new emerging trends of applying AI to diverse fields as well as trying to improve access to AI around the world as well. This proves that AI is really becoming more ubiquitous in our lives.

Networking

During the conference, I was able to interact with other delegates such as Dr Tony Low (Hong Kong University of Science and Technology, Hong Kong) and Mr Daniel Caparro (Polytechnic University of the Philippines & Academia, Philippines) to find out more about their works and research journey.

(From left to right: Dr Tony Low, Mr Daniel Caparro, me)

Reflection

In conclusion, I believe this was an exciting and enriching conference where I was able to interact with other delegates and explore the vast expanses of the world of AI. The presentations were meaningful, well-pitched and in-depth, and the question-and-answer segment also allowed me to better understand others' works.The opportunity to interact with professors and researchers from other countries and to understand works that address very pressing real-world problems was very insightful. I believe this conference was worth it to attend and it was truly a great learning opportunity.

Cited works

All works cited are from IEEE Region 10 Conference 2024, and will be published as part of the conference proceedings.

1. Borines V. A., Turi A. N., & Hedru P. The What, So What, and What Now of the AI Landscape in Emerging Economies.
2. Silapachote P., Srisuphab A., Damkham W., Korkiattrakool P., & Songdechakaivu K. Practical Mobile Based Services for Identification of Chicken Diseases From Fecal Images.
3. Bautista E. O., Casama M. A. D., Nepomuceno D. D., & Pedrasa M. A. Detecting and Localizing Electricity Pilferage Using Deep Neural Networks in a Low Voltage Distribution Network With Both Smart and Traditional Meters.
4. Ercan F. & Mathyalakan A. Computer Vision Based Safety Inspection in Public Housing.
5. Abudin M. D. J., Thokchom S., & Panda G. Data Interpolation for Mitigating False Data Injection Attacks in Smart Grids.
6. Yoshida T., Uruma K., & Sasaki R. GAN-Based Tone Curve Learning Approach to HDR Image Reconstruction Method Using Single LDR Image.
7. Paul D., Chen J., Yin Y., Wijaya C., Yamaura S., Kumar S., Miura K., Tajika Y., & Higuchi S. Cyber Threat Analysis Through Densified Causal Knowledge Graph.